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CSE 445 – MACHINE LEARNING PROJECT

Image Super-Resolution Using a Deep Learning-Based Model (SRCNN)

Individual Progress Report – Member 2

1. Introduction

This report summarizes the implementation and evaluation work completed for the image super-resolution project. The assigned responsibilities include preparing image pairs, establishing a Bicubic interpolation baseline, implementing and training SRCNN models, and comparing reconstruction quality using PSNR and SSIM.

2. Dataset Preparation

The provided high-resolution (HR) image folder contained 50 JPG images. Matching filenames were available in the provided low-resolution (LR) folder, but the exact procedure used to generate those supplied LR images was not documented in the README. For the experiments reported here, a separate synthetic x2 dataset was created from the HR images: each image was converted to RGB, cropped to even dimensions, downscaled to half width and half height using Bicubic interpolation, and then upscaled to its original dimensions using Bicubic interpolation. This produced 50 synthetic LR/HR pairs. The supplied LR folder was not used to train or evaluate the reported models.

3. Methodology

3.1 Bicubic Baseline

Bicubic interpolation was used as the baseline reconstruction method. In the synthetic dataset, the LR input was upscaled back to the original HR dimensions using Bicubic interpolation. The resulting images were compared with their HR targets.

3.2 SRCNN Implementation and Training

A three-layer SRCNN architecture was implemented using convolutional layers with 64, 32, and 3 output channels, kernel sizes of 9, 1, and 5, and ReLU activations between the convolutional layers. The model was trained using Mean Squared Error (MSE) loss and the Adam optimizer. The dataset was split into 40 training images and 10 validation images using a fixed random seed. Training used 64 × 64 patches.

3.3 Fine-Tuning and Residual SRCNN

The initial SRCNN was fine-tuned with a lower learning rate. A residual variant was then trained to predict a correction to the Bicubic-upscaled input, using the form output = input + learned residual. The residual model was trained for 40 epochs with a learning rate of 0.0001. Its best recorded validation loss was 0.003802, and the checkpoint was saved as residual_srcnn.pth in the Colab runtime.

4. Evaluation Metrics

Peak Signal-to-Noise Ratio (PSNR) measures pixel-level reconstruction error, with higher values generally indicating better reconstruction. Structural Similarity Index Measure (SSIM) compares structural similarity, with values closer to 1 generally indicating greater similarity. The reported numbers are averages over the same 10 validation images.

5. Results

Method	**Average PSNR (dB)**	**Average SSIM**
Bicubic interpolation	27.2226	0.8006
Fine-tuned SRCNN	25.4545	0.7505
Residual SRCNN	27.5858	0.8167

The Residual SRCNN achieved the best average scores on the validation set. Compared with Bicubic interpolation, its PSNR increased by 0.3632 dB and its SSIM increased by 0.0161. The fine-tuned SRCNN did not outperform the Bicubic baseline on these metrics. These results show that the residual formulation improved performance for this specific experimental setup, but they do not establish that it will outperform Bicubic interpolation on all images.

6. Limitations

The evaluation set contains only 10 validation images, so the results may not generalize to larger or different datasets.

The experiments use synthetically generated Bicubic degradation rather than the supplied LR images or independently captured low-resolution images.

The visual comparison should be created from actual model outputs. A conceptual infographic should not be presented as an actual model-output comparison.

The best validation loss was selected using validation patches, while the reported PSNR and SSIM were calculated on full validation images.

7. Conclusion

The assigned implementation and evaluation work was completed in Google Colab and the notebook was committed to the team GitHub repository. Among the tested methods, Residual SRCNN achieved the highest validation PSNR and SSIM. Future work should evaluate the model on the supplied LR folder after confirming how those images were generated, produce a side-by-side comparison from actual predictions, and test on a larger validation set.

8. Reproducibility Notes

The notebook contains the experimental code. The reported setup used 50 HR images, a 40/10 train-validation split, 64 × 64 training patches, MSE loss, and the Adam optimizer. The residual model checkpoint was saved at /content/residual_srcnn.pth during the Colab session. Colab runtime files may be temporary, so important checkpoints should be downloaded or saved to Google Drive if they are needed later.